

No. of Functional Features Included in the Solution

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	5 Marks

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)
1	Single-applicant prediction form	Analyst/customer enters one applicant's details and gets an instant decision	Flask /predict route + index.html	Done
2	Decision stamp + confidence score	Shows Approve/Decline with a calibrated probability	result.html	Done
3	JSON API endpoint	Programmatic access to the same prediction logic	Flask /api/predict route	Done
4	Batch CSV screening	Compliance officers upload a CSV and screen many applicants at once	Flask /batch/predict route + batch.html	Done
5	Downloadable batch results	Full scored CSV available to download	Flask /batch/download route	Done
6	4-model comparison & selection	Trains and compares Logistic Regression, Decision Tree, Random Forest, XGBoost; auto-selects the best by ROC-AUC	data/train_models.py	Done
7	Calibrated decision threshold	Avoids over-penalizing borderline-good applicants vs. a raw 0.5 cutoff	data/train_models.py + app.py	Done
8	Model transparency report	Shows comparison metrics, ROC curves, and feature importance	Flask /about route + about.html	Done

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (prediction form, stamp+confidence, batch screening, downloadable results, model comparison, calibrated threshold)
Additional / Nice-to-Have Features	2 (JSON API, model transparency report)
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	New application form, batch upload page, model report page
2	Backend / Logic	3	Feature engineering, model scoring, decision calibration
3	Database / Storage	1	CSV-based application & credit history data, saved model artifacts
4	API / Integration	1	JSON API endpoint (and optional IBM Watson ML endpoint)
5	Security / Authentication	0	Not yet implemented — see Future Scope